SHELJIN S.A

Embedded AI | Robotics | IoT | Backend Engineer

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PROFESSIONAL SUMMARY

Embedded AI and Robotics Engineer with 4+ years of experience designing intelligent robotic systems, embedded firmware, IoT platforms, and scalable backend applications. Experienced in Raspberry Pi, ROS 2, FastAPI, Django, MQTT, Computer Vision, and autonomous systems. Passionate about building production-grade embedded software, robotics platforms, and edge AI solutions.

WORK EXPERIENCE | 4 Years

IoT Engineer | Foxtech Private Solutions — 03/2025 - Present

AI Rover

- Built an autonomous robotic system using Raspberry Pi 5, running two AI models concurrently for object detection and classification, integrating sensor controls and optimized with multi-threading for real-time performance.
- Implemented multiple autonomous modes with real-time live camera feedback, IMU sensor integration, and compass data for precise orientation, supporting full offline operation.
- Managed device control and sensor data using threading and FastAPI endpoints with async/await, achieving sub-100 ms response latency.
- Optimized navigation algorithms to improve obstacle avoidance by 30%, ensuring seamless hardware-software integration with motors, sensors, and IoT modules.

ROS 2 Mapping

- Developed a 3D environment mapping system using LiDAR sensors and ROS 2, creating real-time point cloud models for autonomous navigation and obstacle detection.
- Implemented SLAM algorithms for precise localization and 3D reconstruction, achieving sub-5 cm spatial accuracy.
- Simulated and tested in Gazebo, optimizing data processing pipelines for low-latency mapping and scalable integration with robotics platforms.

IoT & Python/Django Developer | Clovion Tech Solutions Pvt., Ltd. — 09/2022 – 02/2025 (2 Years 6 Months)

Smart Bike Lock System

- Designed and developed an IoT-enabled smart bike locking system integrating embedded hardware, backend services, and a mobile application for secure remote access.
- Developed backend services and REST APIs for real-time communication between the mobile application and IoT devices.
- Integrated Arduino, Bluetooth, and embedded controllers for secure lock/unlock operations.

E-Commerce API

- Designed and implemented scalable RESTful APIs using DRF, Swagger UI, Celery, Redis, 2FA, JWT, and Eraser.io.
- Optimized backend performance by 25%, improving system stability and response times.
- Collaborated within a 5-member agile team, focusing on microservices, asynchronous tasks, and real-time API design using WebSockets.
- Collaborated with cross-functional teams to deliver a production-ready IoT solution for client deployment.

Billing Software

- Built a high-performance billing system using Django, ReportLab, and Celery for automated invoice generation.
- Integrated MySQL with ORM and Redis caching to boost query efficiency by 30%.
- Focused on deployment, CI/CD, and reliability using GitHub Actions and cPanel, ensuring version consistency across multiple environments.

Hardware Deployment & Management

- Deployed DRF APIs, IoT MQTT brokers, and static web apps across cPanel and cloud environments.
- Managed MySQL and Redis databases, automated background tasks using Celery, and reduced REST API latency by 20%.
- Maintained GitHub-based version control and streamlined CI/CD pipelines for scalable, production-ready systems.

PERSONAL PROJECTS

Sound or Mode Controlled - https://github.com/ShelJi/nano-soundcontrol-pwm-manager

- Developed a sound or mode-controlled automation system using Arduino Nano, PWM, and utime/microtime methods for precise signal timing.
- Integrated a microphone sensor and multiple lighting control modes, achieving a 25% improvement in reaction accuracy and enhanced real-time responsiveness through optimized interrupt handling and hardware-level tuning.

Bike Locker - https://github.com/ShelJi/bike_lock

- Implemented mobile control to enhance user interaction and improved response time by 30%.

Web Scrapper Image Download – https://web-scrapper-image-downloader.streamlit.app/

- Built with Python, Streamlit, and Playwright to extract 1000+ high-quality images from Unsplash, including premium, watermark-free content in maximum resolution via automated scraping.

Facial Attendance System - https://github.com/ShelJi/admin-panel

- Developed a real-time facial recognition attendance system using Django, OpenCV, and NumPy, integrating IoT cameras for live authentication and automated record logging.
- Enhanced system accuracy by 15% through optimized image preprocessing and implemented secure database management for reliable attendance tracking.

Battery Management - https://github.com/ShelJi/battery_manager

Plant-Moisture Data Analyzer - https://github.com/ShelJi/-raspberry_pi_pico_w_Thingspeak_DHT

SKILLS

Programming Languages: Python, C, C++, JavaScript, SQL, HTML, CSS, Bash

Embedded Systems & IoT: Raspberry Pi, Raspberry Pi Pico, Arduino, ESP32, ESP8266, MicroPython, Embedded Linux, GPIO, PWM, ADC, Interrupts, Timers, Sensor Interfacing

Robotics & Autonomous Systems: ROS 2, Gazebo, RViz, SLAM, Sensor Fusion, LiDAR, IMU, Autonomous Navigation, Computer Vision, AI Rover, Mission Planner, MAVLink, Drone Systems

Communication Protocols: MQTT, HiveMQ Cloud, Eclipse Mosquitto, TCP/IP, HTTP/HTTPS, WebSocket, SSL/TLS, UART, SPI, I²C, CAN, Ethernet, Wi-Fi, Bluetooth, RS-232

Backend Development: Django, Django REST Framework (DRF), FastAPI, REST APIs, JWT Authentication, Celery, Redis, Swagger/OpenAPI, WebSockets

Frontend: React.js, HTMX, Tailwind CSS, Bootstrap, GSAP

AI & Computer Vision: YOLO, OpenCV, NumPy, Pandas

Databases: MySQL, PostgreSQL, SQLite, Redis

Operating Systems: Linux, Ubuntu, AlmaLinux, Raspberry Pi OS (Raspbian), Windows

Cloud, DevOps & Infrastructure: AWS, Docker, Kubernetes (K3s), Git, GitHub, GitHub Actions, CI/CD, Nginx, cPanel

Tools & Platforms: Postman, Streamlit, Selenium, ReportLab, Eraser.io, ThingSpeak, Blynk, Mission Planner

Core Competencies: Embedded Systems, Robotics, Internet of Things (IoT), Edge AI, Real-Time Systems, Embedded Firmware Development, Hardware–Software Integration, Sensor Integration, Device-to-Cloud Communication, Asynchronous Programming, Multithreading, Microservices Architecture, System Design

COMPETITIVE CODING PROFILE

- **CodeChef** - codechef.com/users/sheljin
- **Leetcode** - leetcode.com/u/shelj/

EDUCATION

Bachelor of Engineering in Electrical and Electronics 05/2020 - 06/2024
Mar Ephraem College of Engineering and Technology Elavuvilai

INTERESTS

- Surf Tech on YouTube, Football, IOT, Problem solving, Electrical, Craft works, Servers (NAS)